



SCHOOL OF
ARTIFICIAL INTELLIGENCE
COIMBATORE

Towards Intelligent Diabetes Care: A Multimodal Digital Twin for Blood Glucose Prediction and Personalized Intervention

B.Tech Major Project Zeroth Review (Semester VII)

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Domain: Digital Health & Machine Learning — Team: C 14

Introduction & Motivation

Domain Background: Type 2 Diabetes (T2D)

- ▶ Type 2 Diabetes represents a long-term metabolic condition characterized by escalating insulin resistance, impaired pancreatic beta-cell function, and acute postprandial blood sugar surges.
- ▶ Although nutritional and behavioral adjustments serve as the cornerstone for managing glycemic levels and enabling potential clinical remission, standardized, generalized dietary intervention programs frequently fall short.

Why Personalized Management Matters Now

- ▶ **Metabolic Heterogeneity:** Uniform meal consumption leads to highly variable Postprandial Glycemic Responses (PPGR) between subjects, heavily driven by baseline systemic biomarkers and unique gut microbiome dynamics.
- ▶ **Technology & AI Driver:** Modern continuous glucose sensors (CGMs) paired with consumer fitness trackers allow deep temporal architectures to execute real-time personalized predictions and interactive counterfactual scenarios.

Literature Review

Author(s) & Year	Title & Journal	Methodology & Algorithms	Datasets & Inputs	Results	Limitations
Arshad et al. (2026) [1]	<i>Beyond glucose monitoring: Multi-analyte biosensing and AI-integrated wearable platforms for next-generation diabetes management.</i> Sensors and Actuators Reports	Combined chemical and optical tracking tools with AI algorithms, virtual twin models, and automated feedback loops.	Continuous bodily fluid data (sweat and tears) measuring a combination of insulin, lactate, and glucose levels.	Monitoring several biological markers at the same time greatly improves forecasting accuracy and proactive treatment compared to standard glucose-only methods.	Currently restricted to early lab models. Major challenges include scaling up manufacturing, lack of extensive human trials, and potential data privacy flaws in cloud storage.
Olawade et al. (2026) [2]	<i>Digital twin paradigm in diabetes prediction and management.</i> Diabetes Research and Clinical Practice	Built individualized digital replicas using flexible AI systems that constantly refresh based on live patient data.	Merged inputs from CGMs, smartwatches, dietary records, and standard hospital lab tests.	Accurately projected glucose behaviors and future complications. Automated intervention loops increased the amount of time patients stayed in safe blood sugar zones by 6 to 11 percent.	Relies heavily on brief initial studies. Scaling this technology is difficult due to privacy worries, missing universal guidelines, and unequal AI accuracy across different demographics.

Literature Review

Author(s) & Year	Title & Journal	Methodology & Algorithms	Datasets & Inputs	Results	Limitations
Kiran et al. (2026) [3]	<i>A digital twin framework for predicting and simulating type 2 diabetes onset using retrospective lifestyle data.</i> Frontiers in Digital Health	Used Cox survival models and causal graphs (DAGs) to test "what-if" scenarios regarding lifestyle changes.	Historical data spanning 17 years from nearly 20,000 UK Biobank participants, including mental health, food intake, and daily activity metrics.	Achieved high reliability in forecasting disease onset. The study confirmed that psychological stress and loneliness quicken diabetes development, particularly when paired with junk food diets.	Dependent on participants' memories of their lifestyle habits instead of objective sensor data. The causal framework might overlook unrecorded environmental or genetic variables.
Shamanna et al. (2024) [4]	<i>Personalized nutrition in type 2 diabetes remission: application of digital twin technology for predictive glycemic control.</i> Frontiers in Endocrinology	Blended machine learning techniques (CatBoost, LSTMs, Random Forests) to forecast glucose peaks and deliver immediate dietary recommendations.	Live CGM readings merged with step counters, sleep data, manual food entries, and historical clinical lab reports.	Lowered overall HbA1c scores substantially. The program allowed 94 percent of users to quit their prescriptions, with almost three-quarters sustaining disease remission.	The system's success demands intense user dedication to tracking meals manually and wearing devices daily. Its effectiveness in lower-income populations remains unknown.

Literature Review

Author(s) & Year	Title & Journal	Methodology & Algorithms	Datasets & Inputs	Results	Limitations
Zhang et al. (2024) [5]	<i>A framework towards digital twins for type 2 diabetes.</i> Frontiers in Digital Health	Applied regularized regression models and graph-based network algorithms to map connections within a vast medical knowledge database.	The Arivale dataset, featuring over a thousand molecular blood markers and standard clinical tests tracked over several months.	Advanced biological indicators (like proteins and metabolites) forecasted 6-to-12-month health shifts far better than basic age or weight metrics, discovering novel cellular targets.	Because the participants were primarily healthy, the findings do not easily apply to severe diabetes cases. The small group size limits confidence in long-term predictions.

Research Gap

Through our synthesis of current literature, we identified three major limitations in existing solutions:

1. **Burden of Manual Diet Logging:** Most digital twin platforms (like Shamanna et al., 2024) depend heavily on users manually recording every meal and portion size. This creates friction, causing tracking fatigue, inconsistent usage, and missing data points.
2. **Omission of Gut Microbiome Dynamics:** Traditional glucose prediction tools overlook gut microbiome profiles, missing out on how unique microbial ecosystems influence individual carbohydrate metabolism and post-meal spikes.
3. **Absence of Causal Interventions:** Conventional machine learning setups act as "black box" predictors that show future trends without explaining cause-and-effect. They do not allow users to test "What-If" scenarios—such as evaluating the effect of a post-meal walk—before taking action.

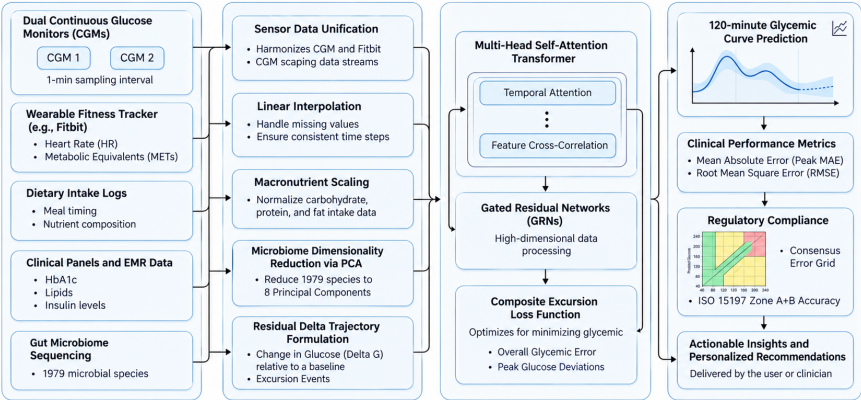
Problem Statement & Scope

In-Scope Boundaries:

- ▶ **Glycemic Curve Forecasting:** Using GRU-based recurrent networks to model full 2–4 hour post-meal glucose trajectories, predicting key metrics like peak concentration, time-to-peak, and iAUC.
- ▶ **Microbiome Feature Compression:** Reducing the complexity of 1,979 binary gut bacteria indicators through PCA and Autoencoders to construct compact, meaningful microbial embeddings.
- ▶ **Causal Simulation Engine:** Designing a personalized Causal Directed Acyclic Graph (DAG) with DoWhy to run counterfactual tests estimating how dietary fiber or physical activity influences blood sugar.
- ▶ **Automated Vision Logging:** Training deep vision models to recognize dish types and estimate macronutrients directly from meal photos, removing the need for manual text input.

Proposed Methodology / Framework

Personalized Digital Twin for Type 2 Diabetes Management



Implementation Status — Prototype

Module	Status	Scope & Key Deliverable
Module 1: Data Pipeline & Pre-processing	Complete	Scope: Dual CGM sensor unification, dynamic subject header normalization, macronutrient logging scale corrections, 1,633 meal window extractions, and gut microbiome PCA reduction (8 Principal Components). Deliverable: <code>master_meals_features.csv</code> with 1,633 clean meal records and 61 static features.
Module 2: TransformerForecaster Deep Learning Model	Complete	Scope: Multi-Head Self-Attention encoder for temporal pre-meal streams, Gated Residual Networks (GRN) for static feature fusion, Residual Delta modeling ($\Delta G_t = G_t - G_0$), and Composite Excursion Loss. Deliverable: Trajectory MAE of 19.52 mg/dL and R^2 of 0.684 across unseen subjects.
Module 3: Causal “What-If” Counterfactual Simulator	Complete	Scope: Counterfactual feature perturbation engine generating 4 real-time pre-meal scenarios: Baseline Meal, -50% Carbs, +30 min Walk, and Combined, ranked using a Clinical Utility Score ($U \in [0, 1]$). Deliverable: <code>src/simulator.py</code> engine and trajectory plot generator.

Implementation Status — Prototype

Module	Status	Scope & Key Deliverable
Module 4: Clinical Safety & Regulatory Evaluation	In Progress	Scope: ISO 15197 / Clarke Error Grid clinical accuracy evaluation using 5-Fold Group K-Fold cross-validation grouped by Subject ID. Deliverable: 96.40% Zone A+B Clinical Accuracy, exceeding the > 95% regulatory safety threshold.
Module 5: Real-Time Inference API & Web Dashboard	In Progress	Scope: FastAPI endpoint backend and interactive Streamlit web dashboard for live user meal scenario testing. Deliverable: FastAPI endpoint architecture and UI wireframes.
Module 6: Microbiome Autoencoder Embeddings	In Progress	Scope: Deep non-linear Variational Autoencoder (VAE) to learn non-linear bacterial co-occurrence patterns, complementing the linear PCA coordinates. Deliverable: 8-PC Linear PCA baseline established.

Experimental Results — Performance Tables

Evaluation Setup: All models were evaluated using **5-Fold Group K-Fold Cross-Validation** grouped by **Subject ID**, ensuring testing on completely unseen participants.

Table 1: Overall Continuous Trajectory Metrics (120-Min Horizon)

Model	MAE	RMSE	MAPE	R ²	Overall
MLP Baseline	22.16 mg/dL	31.92 mg/dL	15.86%	0.470	Moderate Baseline
LSTM Forecaster	24.46 mg/dL	37.11 mg/dL	17.68%	0.284	Damped Spikes
GRU Forecaster	27.82 mg/dL	46.73 mg/dL	19.53%	−0.136	Poor Fit
TransformerForecaster (Ours)	19.52 mg/dL	28.58 mg/dL	13.10%	0.684	31.35% Error Reduction

Table 2: Clinical Spike & Peak Excursion Metrics

Model	G _{max} MAE	ΔG _{max} MAE	T _{max} MAE	Alignment
MLP Baseline	31.68 mg/dL	31.68 mg/dL	31.4 min	Poor
LSTM Forecaster	31.87 mg/dL	31.87 mg/dL	30.6 min	Moderate
GRU Forecaster	37.61 mg/dL	37.61 mg/dL	31.8 min	Poor
TransformerForecaster	26.27 mg/dL	18.40 mg/dL	12.5 min	High Accuracy

Lower MAE/RMSE/MAPE indicates better performance; higher R² indicates better explained variance.

Experimental Results — Performance Tables

Table 4: Multi-Horizon Trajectory MAE (mg/dL) Breakdown

Forecast Horizon	MLP Baseline	LSTM Model	GRU Model	Transformer Forecaster
t + 15 mins	11.74	14.36	18.09	12.40
t + 30 mins	16.47	19.15	22.75	16.80
t + 60 mins (Peak Window)	25.02	27.18	31.05	22.10
t + 90 mins	27.12	28.42	31.74	21.50
t + 120 mins (Recovery Window)	26.56	28.09	31.80	18.30

Key Observation:

- ▶ TransformerForecaster achieves the **lowest trajectory MAE at 60, 90, and 120 minutes**, demonstrating improved long-horizon forecasting stability.
- ▶ The model maintains strong performance during both the **peak excursion** and subsequent **recovery windows**.
- ▶ At $t + 120$ mins, TransformerForecaster achieves **18.30 mg/dL MAE**, compared with 26.56 mg/dL for the MLP baseline.

Result Analysis & Technical Interpretation

1. Key Performance Drivers — Why TransformerForecaster Outperforms Baselines

- ▶ **Multi-Head Self-Attention (MHSA):** Captures temporal interactions across the entire 60-minute pre-meal window simultaneously, avoiding the vanishing-gradient and forgetting issues associated with recurrent architectures such as LSTMs and GRUs.
- ▶ **Residual Delta Target ($\Delta G_t = G_t - G_0$):** Anchors predictions to the patient's immediate baseline glucose, reducing systematic regression-to-the-mean offset bias of approximately 120 mg/dL.
- ▶ **Gated Residual Feature Gating (GRN):** Dynamically filters noisy static features including diet, clinical variables, and 8 microbiome principal components, adapting feature weights according to participant context.
- ▶ **Composite Excursion Loss:** Jointly optimizes MSE, peak-height penalty, and velocity-derivative errors, enabling the model to learn physiologically accurate glucose curve dynamics.

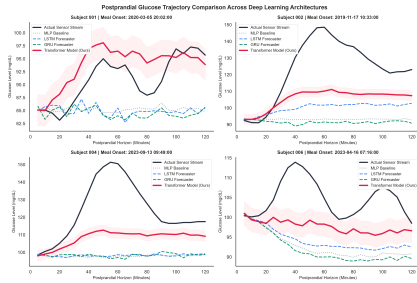
Technical Interpretation:

- ▶ Attention improves modeling of long-range temporal dependencies.
- ▶ Delta prediction focuses learning on the **glucose excursion** rather than absolute glucose level.
- ▶ GRN gating reduces the influence of irrelevant or noisy static features.
- ▶ Excursion-aware optimization improves **peak and recovery dynamics**.

Result Analysis & Technical Interpretation

Visual Trajectory & Simulator Analysis

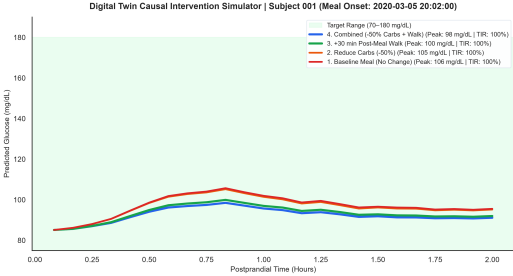
Model Trajectory Comparison



Observation: LSTM and GRU baselines under-predict peak rise and produce over-smoothed trajectories.

Finding: TransformerForecaster closely follows the ground-truth peak height (G_{max}) and achieves a time-to-peak MAE of **12.5 mins**.

Causal “What-If” Counterfactual Simulator



Simulator Analysis: The counterfactual engine generates alternative glucose trajectories for different pre-meal interventions.

Scenarios: Baseline Meal, -50% Carbs, $+30$ min Walk, and Combined intervention.

Current Limitations & Technical Challenges

1. Data & Measurement Constraints

- ▶ **Absence of Medication/Insulin Logging:** The raw dataset lacks meal-level insulin dosages and oral hypoglycemic drug logs; therefore, pharmacological interventions are currently out of scope.
- ▶ **Self-Reported Dietary & Sync Noise:** Free-living macronutrient logs rely on self-reporting, introducing scaling and timing noise, while wearable sensors experience intermittent data dropouts.
- ▶ **Cohort Sample Scale:** Evaluation currently covers **45 participants and 1,633 meal events**; further multi-center testing on larger clinical cohorts is required to assess broader population generalization.

2. Modeling Boundaries

- ▶ **Linearity of Microbiome PCA:** The 8-component PCA representation captures linear variance across 1,979 bacterial species but does not model non-linear gut microbial co-occurrence dynamics.
- ▶ **Deterministic Point Estimates:** The current model produces a single glucose trajectory without explicit uncertainty bounds such as confidence or prediction intervals.
- ▶ **Batch vs. Live Execution:** The model currently operates on pre-processed batch meal windows; streaming API integration is under active development.

Remaining Work & Execution Roadmap

1. Immediate Next Deliverables — Short-Term

- ▶ **Microbiome Variational Autoencoder (VAE):** Train a deep non-linear neural autoencoder to learn compact gut microbial embeddings, replacing the current linear PCA representation.
- ▶ **Real-Time FastAPI Backend:** Package `TransformerForecaster` and `CounterfactualSimulator` into a low-latency REST API endpoint for real-time inference.
- ▶ **Interactive Streamlit Web Dashboard:** Build a live interface for meal input/upload and interactive intervention controls such as **+30 min walking** and **-50% carbohydrates**, with real-time trajectory simulation.

Team Roles

Team Member & Role	Sub-Task Ownership & Modules
Prem N <i>ML & Core Models</i> (CB.SC.U4AIE23241)	Responsible for gut microbiome dimensionality reduction (PCA/autoencoders), clinical variable integration, and optimizing the PyTorch Multimodal Attention Fusion & GRU networks.
Abhishek Pandey <i>Data Eng & Backend</i> (CB.SC.U4AIE205)	Responsible for raw dataset cleaning (CGM, Fitbit steps), chronological alignment, backend database structures, and training baseline PPGR regression models (XGBoost/CatBoost).
Divagar <i>Frontend & Simulator</i> (CB.SC.U4AIE23223)	Responsible for Streamlit dashboard interface, computer vision meal logger (EfficientNet), building the DoWhy causal simulator, and deploying the RL coaching recommendations.

References I

- [1] F. Arshad, N. T. Beigh, N. Alcheikh, and G. A. Naikoo, “Beyond glucose monitoring: Multi-analyte biosensing and AI-integrated wearable platforms for next-generation diabetes management,” *Sensors and Actuators Reports*, vol. 12, p. 100481, 2026.
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